# SOMAIYA

VIDYAVIHAR
K J Somaiya Institute of Technology
An Autonomous Institute permanently affiliated to University of Mumbai

## Department of Artificial Intelligence & Data Science

## MINI PROJECT- II LOG BOOK

## SY SEM III (ODD SEM)

### Academic Year 2026-27

| Roll No. | Name of the Student |
|---|---|
| 1 | Sameera Amberkar |
| 54 | Mahi Satoskar |
| | |

| | |
|---|---|
| **Name of the Project** | IMD Aviation forecast verification web Portal. |
| **Name of Guide** | Prof. Pravin Patil |
| **Name of Co-Guide (If any)** | |

## DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

**Vision, Mission, Program Educational Objectives (PEOs) and Program Specific Outcomes (PSOs) and Program Outcomes (POs)**

## INSTITUTE VISION

To be universally accepted as a synonym of quality, excellence and commitment in the field of engineering education by nurturing talent and **transforming young minds** to realize their potential and become **future ready engineers**.

## INSTITUTE MISSION

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| **Statement 1** | To provide students with a thorough **knowledge of engineering** to refine their professional skills. |
| **Statement 2** | To **nurture creativity and innovation** while encouraging multidisciplinary interaction. |
| **Statement 3** | To train students to be **industry ready and capable of working effectively** as an individual and in a team. |
| **Statement 4** | To **inculcate ethical behavior, responsibility and commitment** among students. |

## DEPARTMENT VISION

To **serve the society** by **transforming young minds** and **preparing them for a bright future.**

## DEPARTMENT MISSION

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| **Statement 1** | To create a holistic educational experience for students focused on subject knowledge, problem solving, inquisitiveness, leadership, communication and team skills. |
| **Statement 2** | To nurture an environment committed to the free exchange of ideas where research, creativity, innovation, and entrepreneurship can flourish. |
| **Statement 3** | To impact society in a transformative way by getting engaged with the external collaborators. |

## PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

After a few years graduates will be able to:

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| **PEO I** | Apply skills required to design and develop solutions applicable in real life situations. |
| **PEO II** | Pursue successful and satisfying careers in the profession of their choice or enterprise. |
| **PEO III** | Pursue higher studies in related fields from recognized institutions. |

## PROGRAM SPECIFIC OUTCOMES (PSOs)

Students of the program will:

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| **PSO I** | Exhibit proficiency in the field of **Artificial Intelligence** by **developing intelligent systems** |
| **PS O II** | Be accomplished in the field of **data analytics, statistical programming languages and Machine learning** |

## PROGRAM OUTCOMES (POs):

The graduates of this program will be able to:

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| **PO_01** Engineering Knowledge (EK): | Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems. |
| **PO_02** Problem Analysis (PA): | Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences. |
| **PO_03** Design/Development of Solutions (DS): | Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations. |
| **PO_04** Conduct Investigations of Complex Problems (IP): | Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. |
| **PO_05** Modern Tool usage (MT): | Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations. |
| **PO_06** The Engineer and The world | Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice. Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. |
| **PO_07** Ethics (Eth): | Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice. |
| **PO_08** Individual and Term Work (ITw): | Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. |
| **PO_09** Communication (Cm): | Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. |
| **PO_10** Project Management and Finance (PM): | Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. |
| **PO_11** Life-long Learning (LL): | Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. |

# SY Artificial Intelligence & Data Science

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| **Semester:IV** | |
| **Subject: Project Based Learning - Minor Project Lab-II Course Code : (AIPR42) Lab Hour: 2 Hrs/week** | **Term work: 25 marks,** |
| **Practical & Oral Exam: 25 marks** | **Total marks= 50 marks** |

## PBL Objectives:

1. To acquaint with the process of identifying the needs and converting it into the problem.
2. To familiarize the process of solving the problem in a group.
3. To acquaint with the process of applying basic engineering fundamentals to attempt solutions to the problems.
4. To inculcate the process of self-learning and research.

## PBL Outcomes:

Learner will be able to…

1. Identify problems based on societal /research needs.
2. Apply Knowledge and skill to solve societal problems in a group.
3. Develop interpersonal skills to work as member of a group or leader.
4. Analyze the impact of solutions in societal and environmental context for sustainable development.
5. Excel in written and oral communication.
6. Demonstrate capabilities of self-learning in a group, which leads to lifelong learning.
7. Demonstrate project management principles during project work

## Guidelines for Activity-Based Learning (PBL):

1. Project is a group activity and students shall form a group of 2 to 3 students. A group shall not be more than three students.
2. Project Based Learning - Minor Project Lab-II should be implemented with hardware and/or software.
3. Students will be assigned an open-ended problem which they will finalize according to their preferences and in consultation with the faculty supervisor.
4. Project should be implementation of Applied Artificial Intelligence/ Data Science/Embedded Systems/ Societal need based / Innovative idea implementation etc.
5. Students shall submit an implementation plan in the form of Gantt/PERT/CPM chart, which will cover weekly activity of mini projects.
6. A collaborative logbook will be prepared by each group, which will be verified regularly by; guide/supervisor can verify and record notes/comments.
7. The solution to be validated with proper justification and report to be compiled in standard format of the college.
8. The focus of project will be on self-learning, innovation, addressing societal problems and based solutions.

# K. J. SOMAIYA INSTITUTE OF TECHNOLOGY

| Assessment Outcome | No Progress (F) | Unacceptable (D) | Basic ( C) | Good (B) | Excellent (A) |
|---|---|---|---|---|---|
| | 0 | 1 point | 2 points | 4points | 5 points |
| Organization | | The report is disorganized to the extent that it prevents understanding of content. Citations are misplaced | The organization is unclear. Documentation and referencing in text are incomplete. | The organization is generally good, but some parts seem out of place. Citations could be better | Written work is well organized and easy to understand. Clear references in proper format. |
| Engineering Theory and analysis | | Little engineering/theoretical analysis is presented. Lots of irrelevant material | Analysis consists of trivial calculations and is poorly explained. | Most of the work presented is relevant to the work performed. | Analysis is presented with sufficient detail to be understood at peer level |
| Spelling and grammar and writing style | | There are frequent misspelled words and serious grammatical errors, indicating that time was not taken to spell-check and proofread. The writing is disorganized, making understanding a difficult task | There is more than one spelling or grammatical error per page. The writing style is readable, but difficult to follow. | There are a few spelling and grammatical errors. The writing style indicates planning that makes reading easy. | The work has been thoroughly spell- check and proofread. The writing style indicates planning that makes reading easy and the flow of material makes understanding easy. |
| Visual, example; graphs/ diagrams | | There are few visual aids, and those used are carelessly prepared. | Most visual aids are sloppy and hard to read. | Visual aids are good, but a few are sloppy or difficult to read. | Visual aids are used frequently. They are easy to read and understand, and are of professional quality. |

## DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

## PROJECT REPORT RUBRIC (10 Marks)

## PROJECT IMPLEMENTATION RUBRIC (10 Marks)

| Assessment Outcome | No Progress (F) | Unacceptable (D) | Basic ( C) | Good (B) | Excellent (A) |
|---|---|---|---|---|---|
| | 0 | 1 point | 2 points | 4points | 5 points |
| Formulation of design problem | | The design problem is not formulated clearly. | The problem formulation is unclear in some respects and does not appear to be well thought out. | The problem formulation is clear, but the scope is not well defined | Design problem formulation is clear and well thought out. The problem scope is well defined |
| Concepts | | Prerequisite course content is not applied correctly. New areas are not included. | Prerequisite course content is used, but new and unfamiliar areas are not introduced. | Prerequisite course content is used easily, and some material beyond the course is included. | Concepts beyond those in the prerequisite course are frequently used. The professor may have learnt something new. |
| Innovation | | No innovative work initiated by students. | Innovative work initiated by student of minimal importance. | Innovative work initiated by student of minimal impact. | Evidence of some promising innovative work initiated by student. |
| Resource Management | | Little evidence of good resource management such as late component sourcing and unplanned late laboratory usage. | Some evidence of resource management but mainly on a reactive basis. | Some evidence of proactive utilization of resources but also some reactive utilization | Evidence of the proactive utilization of resources such as planned laboratory usage and working within the confines of existing EE resources. |
| Team Work | | Absolutely no coordination between the team members and lack of communication. | Poor coordination between team members | Team members work coherently on the project but lack of proper communication. | Excellent coordination and team work displayed. |
| Punctuality | | Always fail to meet the deadlines | Very rarely are on time to show the results | Always keep preparing till the last moment of the deadline. | Always ready with the complete assigned work ahead of the deadline. |

## PROJECT DETAILS

1. **Title of Project:** IMD - Aviation Weather Forecast verification

**In- House** **Industry**

2. **Guide:** Prof. Praveen Patil

**Internal:** _____ **External:** _____

***(ONLY FOR INDUSTRY PROJECT)***

3. **Name and Address:** _____

4. **Industry Detail** ***(ONLY FOR INDUSTRY PROJECT):***

**Name and Address:**

**Phone / Mobile No:** _____

**Department: Artificial Intelligence and Data Science**

**5. Student Details:**

| Sr. No | Students Name | Roll No |
|---|---|---|
| 1 | Sameera Ambekar | 1 |
| ~~54~~ 2 | Mahi Satostar | 54 |
| | | |

Weekly Report

| Sr. no | Week No & Date | Discussion | Guide Sign |
|---|---|---|---|
| | **Week-01**<br>**Date:** 13/7/26<br>**(From:** 13/7/26<br>**To:** 20/7/26 ) | Team formation, And Project topic selection | |
| | **Week-02**<br>**Date:** 20/7/26<br>**(From:** 20/7/26<br>**To:** 27/7/26 ) | Finalised Project topic & gone through research paper | |
| | **Week-03**<br>**Date:** 27/7/26<br>**(From:** 27/7/26<br>**To:** 03/8/26 ) | Made PPt, and made corrections, Finalized 2 page report & Panel Presentation | |
| | **Week-04**<br>**Date:** 07/8/26<br>**(From:** 10/8/26<br>**To:** 17/8/26 ) | Implemented ML related changes. | |
| | **Week-05**<br>**Date:** 14/9/26<br>**(From:** 01/9/26<br>**To:** 5[illegible]9/26 ) | Making of poster, github Push, Research paper. | |
| | **Week-06**<br>**Date:** 16/10/26.<br>**(From:** 30/09/26<br>**To:** 16/09/26 ) | uploading all documents and verifying with Sir. | |

| Sr. no | Week No & Date | Discussion | Guide Sign |
|---|---|---|---|
| | **Week-07**<br>**Date:** 23/10/26<br>**(From:** 19/10/26<br>**To:** 23/10/26 ) | Did changes which were discussed earlier. in PPT | |
| | **Week-08**<br>**Date:** 24/10/26<br>**(From:** 5/10/26<br>**To:** 17/10/26 ) | contacted with industry to confirm if there are any changes further. | |
| | **Week-09**<br>**Date:** 24/10/26<br>**(From:** 19/10/26<br>**To:** 24/10/26 ) | Increased Accuracy in aerodrome warning session. | |
| | **Week-10**<br>**Date:** 31/10/26<br>**(From:** 26/10/26<br>**To:** 31/10/26 ) | Discussed everything before submission. | |
| | **Week-11**<br>**Date:** 6/11/26<br>**(From:** 2/11/26<br>**To:** 6/11/26 ) | Took sign on checkuit & submitted github link. | |
| | **Week-12**<br>**Date:**<br>**(From:**<br>**To:** ) | | |